

# Urban Heat Island & Green Space Equity Analysis

Hyderabad Metropolitan Region, Telangana, India

Technical Report

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## Abstract

This report presents a geospatial analysis of urban heat exposure and green space accessibility across the Hyderabad Metropolitan Region, Telangana, India. Combining Landsat thermal imagery, Sentinel-2 spectral indices, gridded population data, and live OpenStreetMap green space records, the analysis identifies which populations face the greatest combined burden of urban heat and poor green space access. Across a study population of 8,376,303, 63.8% (5,340,189 people) live in areas classified as High or Very High Urban Heat Island intensity. Of these, 2,271,067 people, 27.1% of the total population live in priority intervention zones: areas that are both high-heat and more than one kilometre from the nearest park or green space. A twelve-year land surface temperature time series (2013–2024) was analysed for a long-term trend; after excluding a low-confidence year with insufficient satellite coverage, no statistically significant warming or cooling trend was detected, a result reported transparently as a genuine methodological finding. The analysis is delivered through a public interactive web dashboard and a set of reproducible Python and Google Earth Engine scripts.

## Study Area

The study area covers the Hyderabad Metropolitan Region, Telangana, India, bounded approximately between 78.20°E–78.75°E and 17.15°N–17.65°N. This extent captures the dense urban core of Hyderabad along with its rapidly urbanising periphery, providing a representative cross-section of established city districts, expanding suburban development, and the surrounding rural-urban transition zone. The region was selected for its combination of strong satellite and open data availability, rapid recent urban growth, and direct relevance to the author's professional background.

## Data Sources

All datasets used in this analysis are free and openly accessible, sourced primarily through Google Earth Engine and OpenStreetMap.

| Dataset                                       | Source                             | Resolution | Purpose                  |
|-----------------------------------------------|------------------------------------|------------|--------------------------|
| Landsat 8 / Landsat 9 (Collection 2, Level 2) | USGS via Google Earth Engine       | 30 m       | Land Surface Temperature |
| Sentinel-2 Surface Reflectance (Harmonized)   | Copernicus via Google Earth Engine | 10 m       | NDVI, NDBI, NDWI         |
| ESA WorldCover 2021                           | Google Earth Engine                | 10 m       | Land cover context       |
| WorldPop 2020                                 | Google Earth Engine                | 100 m      | Population density       |
| OpenStreetMap                                 | osmnx (live query)                 | Vector     | Parks and green space    |

# Methodology

## Land Surface Temperature Processing

Landsat 8 (2013–2021) and Landsat 9 (2022–2024) thermal band data were retrieved from Google Earth Engine and cloud-masked using the QA\_PIXEL band, removing pixels affected by cloud and cloud shadow. Radiance values were converted to Land Surface Temperature in degrees Celsius using the standard USGS Collection 2 Level 2 scale and offset factors. A pre-monsoon seasonal window (15 February to 31 May) was used throughout, as this represents the hottest and most inter-annually comparable period in Telangana's climate calendar.

## Urban Heat Island Index

NDVI, NDBI, and NDWI were computed from Sentinel-2 surface reflectance imagery and resampled onto the LST raster grid using average resampling. Water bodies ( $\text{NDWI} > 0.1$ ) were masked prior to analysis to prevent naturally cool water pixels from distorting the heat classification. LST, NDVI, and NDBI were each normalised to a 0–1 range and combined into a single Urban Heat Island score using the formula:  $\text{normalized (LST)} - \text{normalized (NDVI)} + \text{normalized (NDBI)}$ . The resulting continuous score was classified into five intensity classes, Very Low, Low, Moderate, High, and Very High, using Natural Breaks (Jenks) classification, computed via a random sample and applied to the full raster. This method was selected over quantile classification, which forces an artificially even distribution across classes regardless of the underlying data pattern, and would have obscured genuine spatial clustering of heat intensity.

## Trend Analysis

An initial two-period comparison (a single year in 2015 against a single year in 2024) produced a misleading result, showing an apparent  $3.5^{\circ}\text{C}$  cooling trend. Investigation identified that 2015 coincided with one of the most severe heatwaves in recent Indian history, making it an unrepresentative baseline rather than evidence of genuine cooling. Averaging three-year windows around each period reduced but did not eliminate the distortion. The methodology was therefore rebuilt as a twelve-year annual time series (2013–2024), with a single regionwide mean LST value computed directly via Earth Engine's `reduceRegion` function for each year, avoiding the need to export and download full-resolution rasters for this component. A linear regression was fitted to the resulting time series using the `scipy.stats.linregress` function. Given that several years had a low number of usable cloud-free scenes, a sensitivity analysis was conducted, reporting trend statistics both including and excluding any year with fewer than three usable scenes, following standard practice in climate trend analysis.

## Green Space Equity Analysis

Parks, gardens, recreation grounds, and forested areas were retrieved directly from OpenStreetMap using the `osmnx` Python library, returning 1,860 tagged features, of which 1,670 were valid polygon geometries. These were rasterised onto the population grid, and a Euclidean distance transform was computed to determine the distance from every location in the study area to the nearest green space. This distance layer was combined

with the Urban Heat Island classification and WorldPop gridded population data to define priority intervention zones: locations classified as High or Very High heat intensity that are also more than one kilometre from the nearest green space. Population figures within these zones were computed by summing the population raster within the zone mask.

## Web Dashboard Development

A public web application was built using Streamlit to make the findings accessible without requiring GIS software. The application includes an interactive Folium map of the Urban Heat Island classification, static maps for each analysis stage, and interactive Plotly charts of population and trend data. To keep the deployed application lightweight and reliable, all raster processing was performed locally, with only pre-processed lightweight assets, PNG images, CSV summaries, and a reprojected overlay image, included in the deployed application, avoiding heavy geospatial dependencies at deployment time.

## Results

The current-period (2022–2024) Land Surface Temperature analysis found a mean temperature of 43.62°C across the study area, with a range from 24.91°C (predominantly water bodies) to 54.89°C, and a standard deviation of 3.36°C.

### Population by Urban Heat Island Class

| UHI Class | Area (sq km) | % of Study Area | Population | % of Population |
|-----------|--------------|-----------------|------------|-----------------|
| Very Low  | 149.22       | 4.7%            | 216,633    | 2.6%            |
| Low       | 387.83       | 12.2%           | 857,950    | 10.2%           |
| Moderate  | 687.47       | 21.7%           | 1,961,531  | 23.3%           |
| High      | 1,059.89     | 33.5%           | 3,476,341  | 41.4%           |
| Very High | 882.87       | 27.9%           | 1,863,848  | 22.2%           |

Combining the High and Very High classes, 63.8% of the study area's 8,376,303 residents live in zones of significant Urban Heat Island intensity.

### Green Space Equity

The mean distance to the nearest green space across the study area was 3.02 km. Within areas classified as High or Very High heat intensity, 2,271,067 people, 27.1% of the total study population live more than one kilometre from any park or green space, meeting the criteria for a priority intervention zone. These zones cover 1,419.98 sq km, representing 44.8% of the study area's land extent.

### Twelve-Year Land Surface Temperature Trend

| Analysis                   | Years          | Trend (°C/decade) | R <sup>2</sup> | p-value | Result                                            |
|----------------------------|----------------|-------------------|----------------|---------|---------------------------------------------------|
| All years                  | 12 (2013–2024) | –3.68             | 0.349          | 0.043   | Significant (likely low-confidence-year artifact) |
| High-confidence years only | 11 (2014–2024) | –2.00             | 0.163          | 0.219   | Not statistically significant                     |

The all-years trend was statistically significant but driven substantially by a single year (2013) with only two usable cloud-free satellite scenes. Excluding this low-confidence year, no statistically significant temperature trend was found in either direction over the 2014–2024 period. This is reported as the primary trend finding, in preference to the more dramatic but less reliable all-years result.

## Discussion

The spatial results indicate that urban heat exposure in Hyderabad is not evenly distributed: nearly two-thirds of the study population lives in zones of significant heat intensity, and just over a quarter live in areas combining high heat with meaningfully restricted access to green space. This pattern is consistent with rapid, uneven urban densification typical of expanding Indian metropolitan regions, where impervious surface growth in peripheral and infill areas has outpaced corresponding green infrastructure investment.

The absence of a statistically significant multi-year LST trend should not be read as evidence that heat exposure is unchanging in Hyderabad; rather, it reflects the genuine limits of what a twelve-year satellite-derived time series can detect once year-to-year weather variability (including strong El Niño-associated heatwave years such as 2015) is properly accounted for. Detecting a robust long-term trend would likely require a longer observation record, denser cloud-free scene coverage per year, or a complementary ground-station temperature record.

## Limitations

Land Surface Temperature, as derived from thermal satellite bands, measures the temperature of the land surface itself rather than near-surface air temperature as experienced by people; the two are correlated but not identical, and LST-based classifications should be interpreted as a proxy for heat exposure rather than a direct measurement of ambient conditions.

The green space equity analysis relies on OpenStreetMap, which has variable completeness depending on how actively an area has been mapped by contributors; some smaller or informally maintained green spaces may be under-represented, which would tend to overstate rather than understate the extent of green space deprivation identified.

Population estimates are drawn from WorldPop's 2020 gridded dataset; more recent population distribution, particularly in rapidly developing peripheral areas, may differ from the figures used here.

## Conclusion

This analysis demonstrates a reproducible, end-to-end geospatial workflow from cloud-based satellite data processing through statistical validation to public dashboard delivery applied to a genuine urban climate equity question. The finding that 2.27 million Hyderabad residents live in areas combining significant heat exposure with poor green space access provides some concrete, spatially explicit basis for prioritising climate adaptation investment, such as targeted tree planting, park development, and cool-surface interventions, in the specific neighbourhoods where the need is greatest.

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